Semantic Analysis: From Information Extraction to Semantic Role Labeling

Information Extraction

- **Definition:** Information extraction is the process of turning unstructured information embedded in texts into structured information (example: relational databases)

Neural Information Extraction

- The simplest approach is to assume relations come from a closed set and to predict the relation given a sentence, where the subject and object are replaced with their NER tags
  - Assumes a binary relation

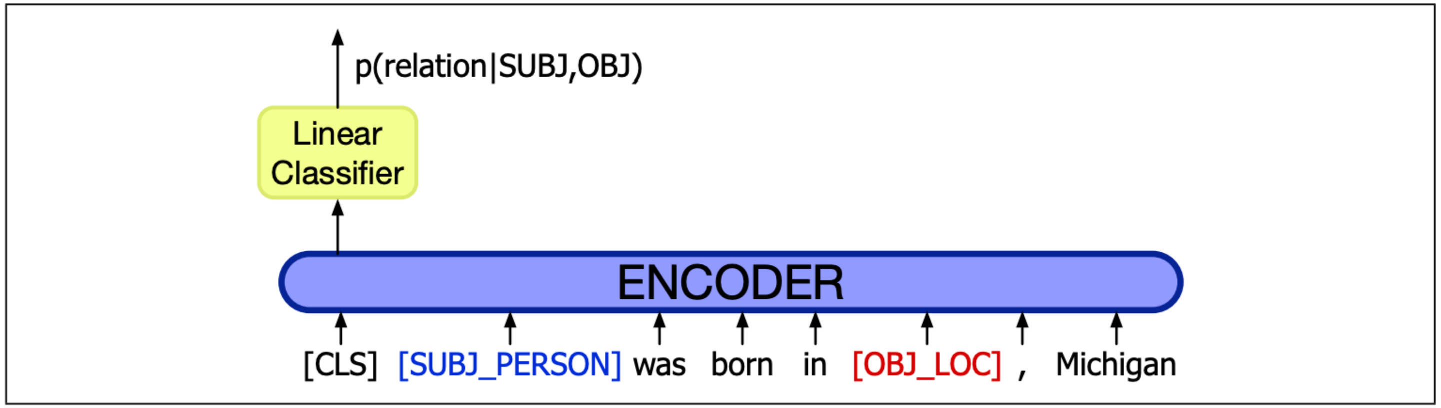


Figure 21.7 Relation extraction as a linear layer on top of an encoder (in this case BERT), with the subject and object entities replaced in the input by their NER tags (Zhang et al. 2017, Joshi et al. 2020).

https://web.stanford.edu/~jurafsky/slp3/21.pdf

Evaluation of Relation Extraction

- Semi-supervised methods are much more difficult to evaluate than supervised methods
  - They extract **new** relations from the web or a large text
  - As methods use very large amounts of text, it is impossible to test them in a sand box with a small pre-annotated gold standard
- Evaluation is therefore done by:
  - Computing precision
  - If the system can rank its output, we can measure precision for different output sizes (e.g., precision for 100 relations, 1000 relations etc.)
  - Extrinsic evaluation by testing how well these relations help, say, question answering

Poon and Domingos, *Unsupervised Semantic Parsing*, EMNLP 2009

Open Information Extraction

- Open Information Extraction (OpenIE) seeks to discover information from plain text
  - One way is to apply syntactic parsing and extract the relations it finds between named entities
  - This works well in some examples:

    “**Pitt** was born in **Shawnee, Oklahoma**”

    “**Jolie** gave birth to daughter **Shiloh Nouvel** in Swakopmund, Namibia, on May 27, 2006.”

  - Extractions:

    born_in(Pitt, Shawnee)

    gave_birth_to_daughter(Jolie,Shiloh_Nouvel)

Open Information Extraction

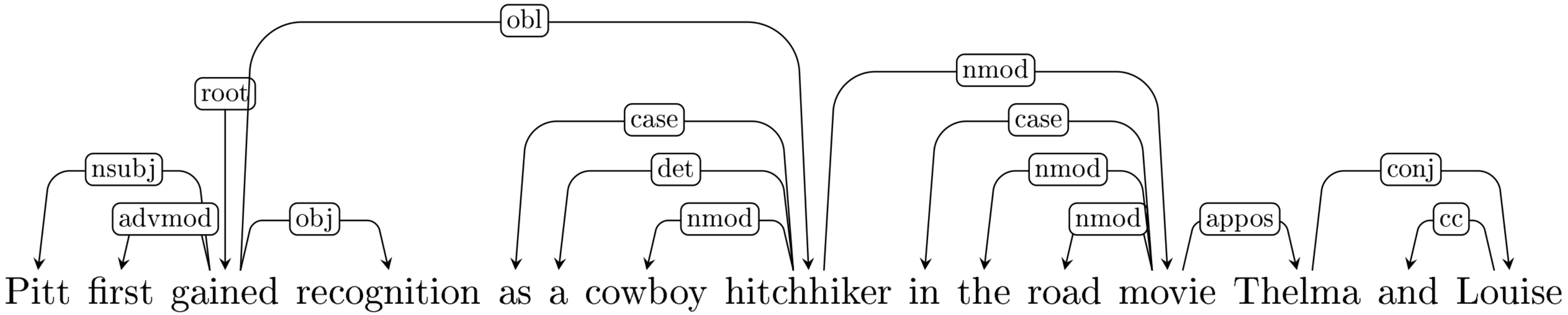
- Open Information Extraction (OpenIE) seeks to discover information from plain text
  - One way is to apply syntactic parsing and extract the relations it finds between named entities
  - Consider this example:

    Pitt first gained recognition as a cowboy hitchhiker in the road movie **Thelma and Louise** (1991). His first leading roles in big-budget productions came with the dramas **A River Runs Through It** (1992) and **Legends of the Fall** (1994), and **Interview with the Vampire** (1994).

  - What's the relation between "Pitt" and "Thelma and Louise"? Between "A River Runs Through It" and "Interview with the Vampire"?

Open Information Extraction

- The dependency path between *Pitt* and *Thelma and Louise* is *nsubj*↑*obl*↓*nmod*↓*appos*



- This is an indirect syntactic relation
  - It would be useful to have a representation that directly encodes the **semantic** relations between the participants